Scenario

In this scenario, you're a security analyst investigating traffic to a website.

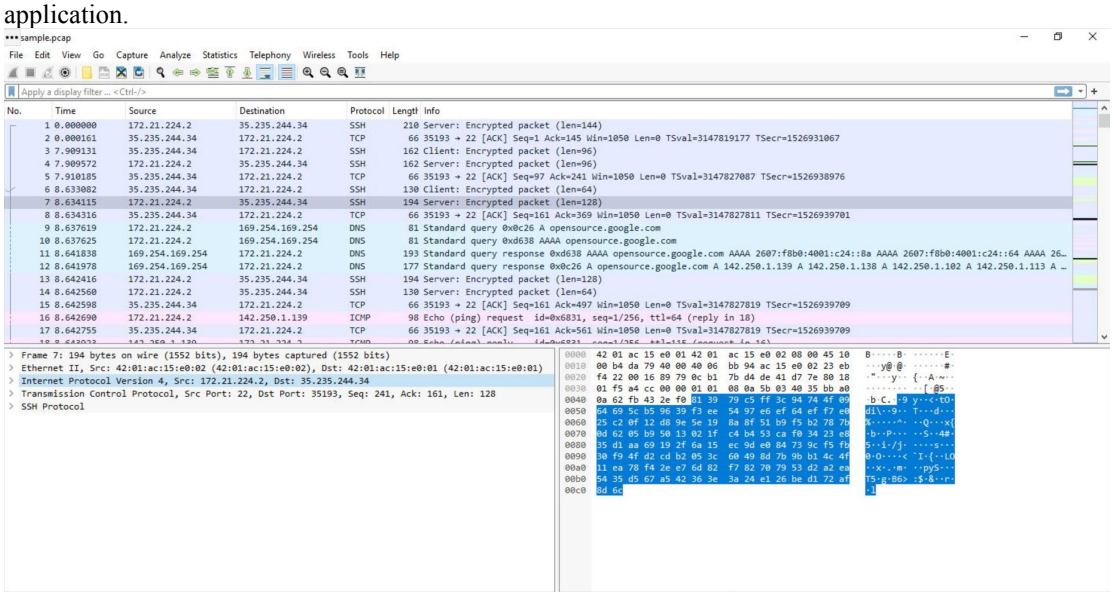
You'll analyze a network packet capture file that contains traffic data related to a user connecting to an internet site. The ability to filter network traffic using packet sniffers to gather relevant information is an essential skill as a security analyst.

You must filter the data in order to:

- identify the source and destination IP addresses involved in this web browsing session,
- examine the protocols that are used when the user makes the connection to the website, and
- analyze some of the data packets to identify the type of information sent and received by the systems that connect to each other when the network data is captured.

Task 1. Explore data with Wireshark

In this task, you must open a network packet capture file that contains data captured from a system that made web requests to a site. You need to open this data with Wireshark to get an overview of how the data is presented in the
application.

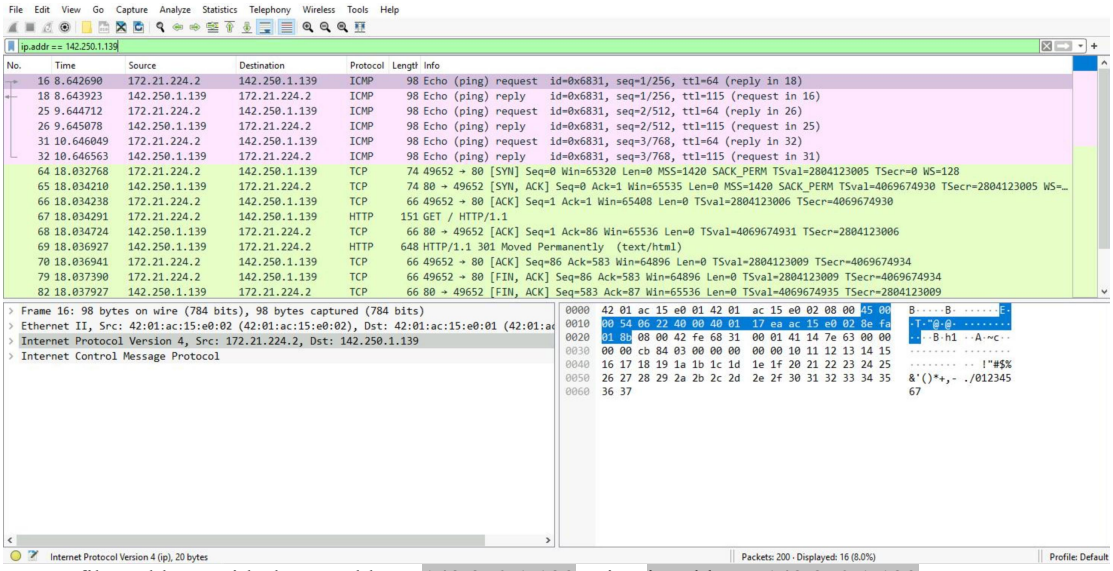


here is an overview of the key property columns listed for each packet:

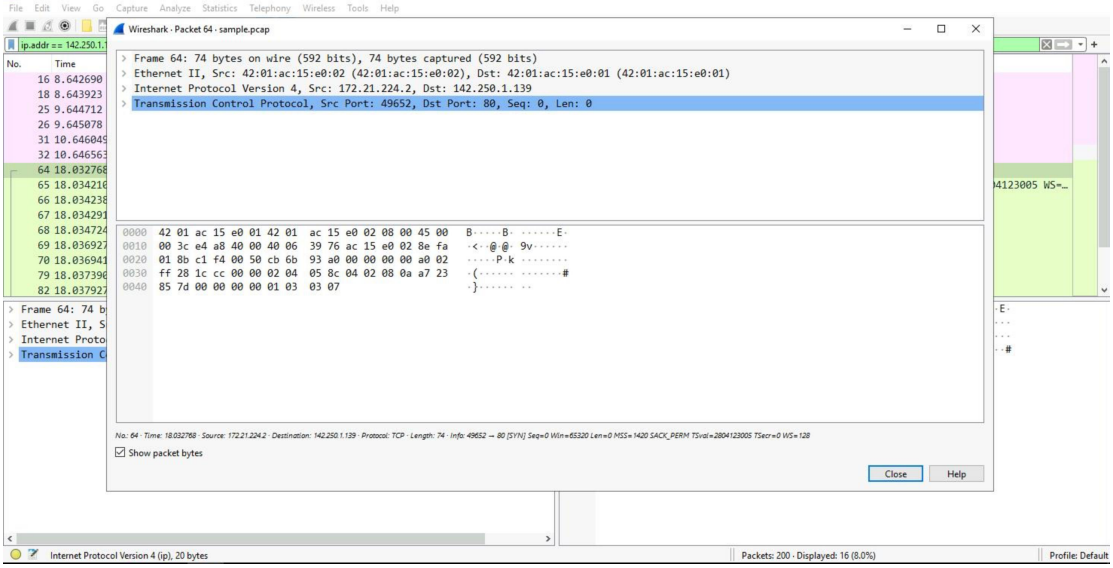
- No. : The index number of the packet in this packet capture file
- Time: The timestamp of the packet
- Source: The source IP address
- Destination: The destination IP address
- Protocol: The protocol contained in the packet
- Length: The total length of the packet
- Info: Some infomation about the data in the packet (the payload) as interpreted by Wireshark

Task 2. Apply a basic Wireshark filter and inspect a packet

In this task, you'll open a packet in Wireshark for more detailed exploration and filter the data to inspect the network layers and protocols contained in the packet.



Here I filtered logs with the IP address 142.250.1.139 using ip.addr == 142.250.1.139
This filter filters all packets with the IP address specified in the expression



Here I opened the packet details window pain. The upper section of this window contains subtrees where Wireshark will provide you with an analysis of the various parts of the network packet. The lower section of the window contains the raw packet data displayed in hexadecimal and ASCII text. There is also placeholder text for fields where the character data does not apply, as indicated by the dot (".")

- The first sub-tree in the upper section provides the details about the overall network packet, or frame, including the frame length and the arrival time of the packet.

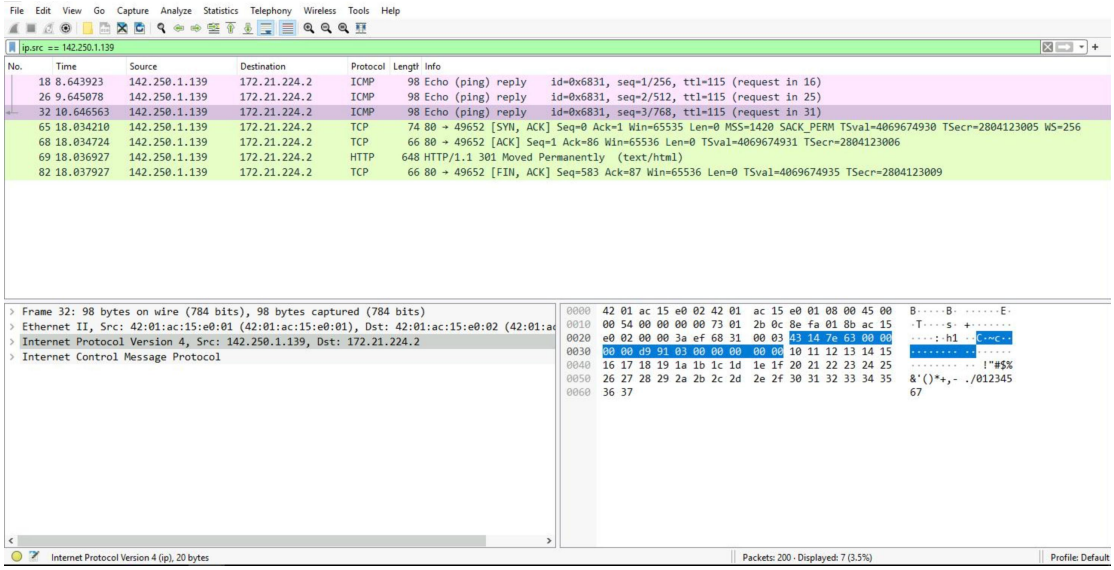
- The Ethernet II sub-tree contains details about the packet at the Ethernet level, including the source and destination MAC addresses and the type of internal protocol that the Ethernet packet contains.

- The Internet Protocol Version 4 subtree provides packet data about the Internet Protocol (IP) data contained in the Ethernet packet. It contains information such as the source and destination IP addresses and the Internal Protocol (for example, TCP or UDP), which is carried inside the IP packet.

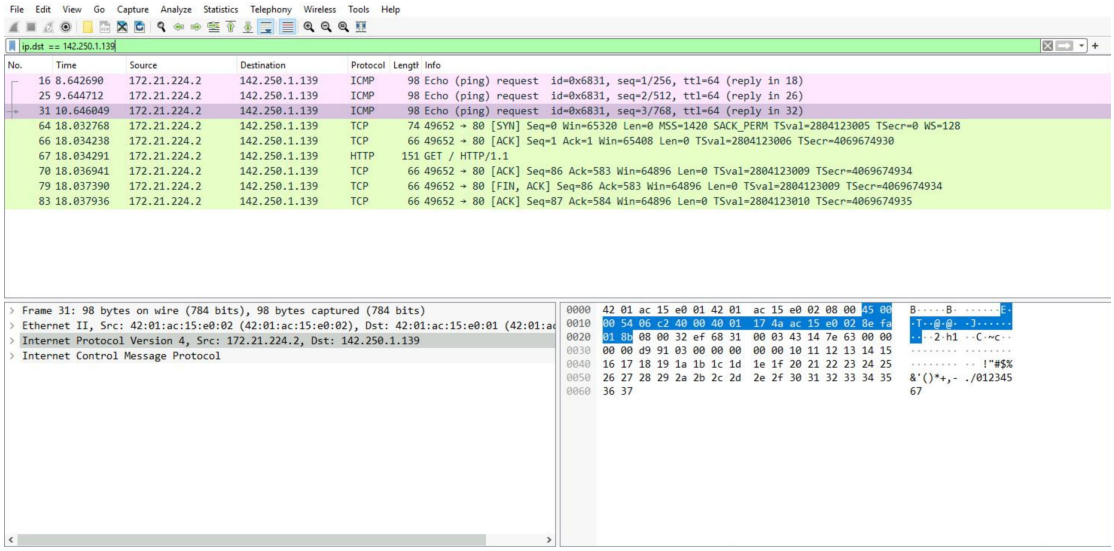
- The Transmission Control Protocol subtree provides detailed information about the TCP packet, including the source and destination TCP ports, the TCP sequence numbers, and the TCP flags.

Task 3. Use filters to select packets

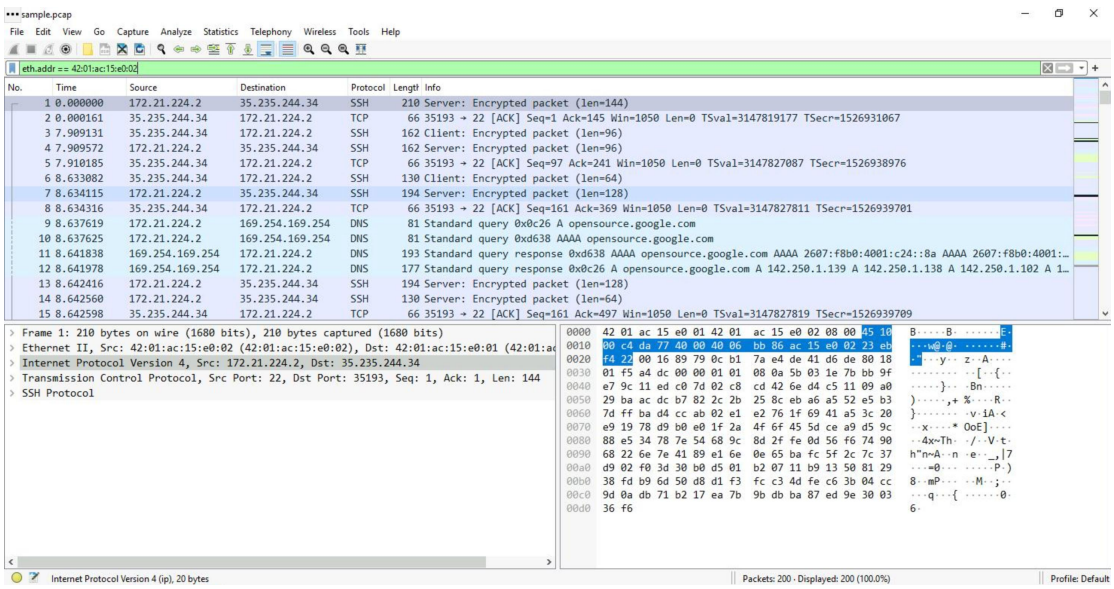
In this task, you'll use filters to analyze specific network packets based on where the packets came from or where they were sent to. You'll explore how to select packets using either their physical Ethernet Media Access Control (MAC) address or their Internet Protocol (IP) address.



In this photo I applied the filter ip.src == 142.250.1.139 this filter applies to all packets with the address specified in the packet here for example im only filtering packets with the source IP address of 142.250.1.139.



In this photo im using ip.dst == 142.250.1.139 to filter all packets with the destination IP address of 142.250.139

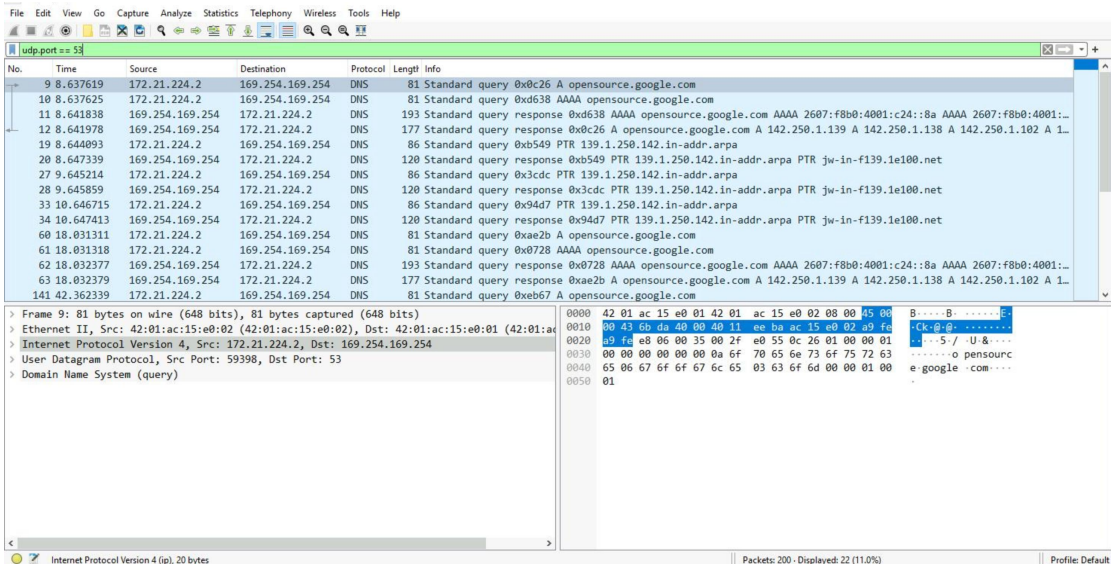


In this photo im using eth.addr == 42:01:ac:15:e0:02 this filters traffic related to one MAC address, regardless of the other protocols involved.

This filter can be helpful especially for a security analyst to find logs related to attacks coming from specific IP addresses. Instead of combing through thousands of logs just to find one single packet we can just apply a filter to quickly find the logs that we need

Task 4. Use filters to explore DNS packets

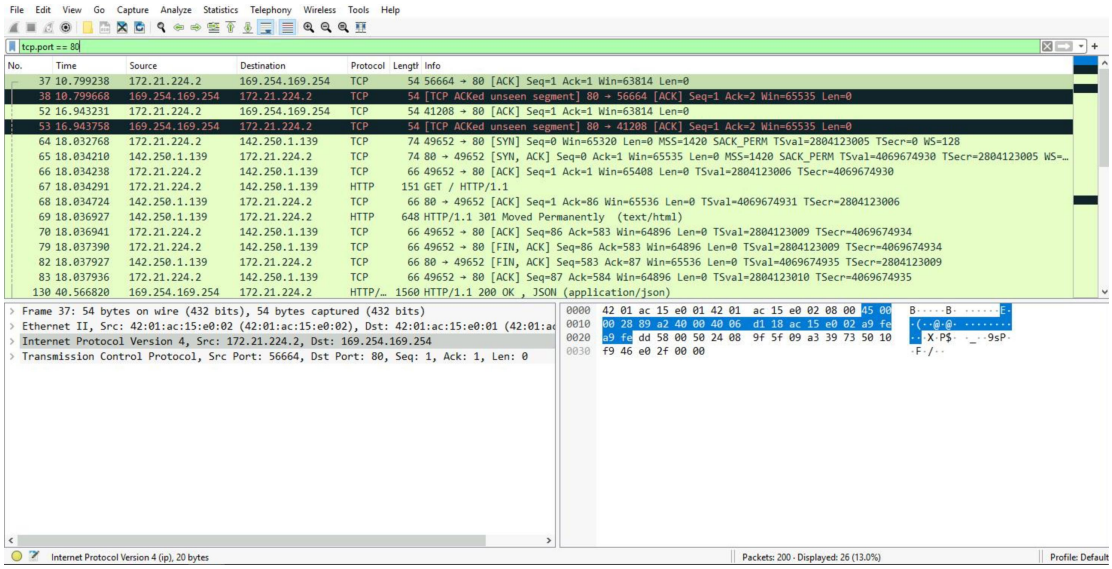
In this task, you'll use filters to select and examine DNS traffic. Once you've selected sample DNS traffic, you'll drill down into the protocol to examine how the DNS packet data contains both queries (names of internet sites that are being looked up) and answers (IP addresses that are being sent back by a DNS server when a name is successfully resolved).



Here I used the filter udp.port == 53 this will list traffic related to DNS queries and responses only.

Task 5. Use filters to explore TCP packets

In this task, you'll use additional filters to select and examine TCP packets. You'll learn how to search for text that is present in payload data contained inside network packets. This will locate packets based on something such as a name or some other text that is of interest to you.



Here I used the filter `tcp.port == 80` this will filter packets using port 80.